

Data sources

This file provides the metadata related to the data sources used for Uruguay in the Human Multiple Births Database (<https://www.twinbirths.org/en/data-metadata/>).

Summary

Data from 1981 to 2020:

Source: Ministerio de Salud Pública – Instituto Nacional de Estadística (MSP-INE).

Coverage and completeness:

- Data include all births that occurred in Uruguay and are registered in the civil registry. Data do not include births registered in Uruguayan consulates abroad.
- Data up until 2007 come from paper-based certificates of live births. Gradual implementation of electronic certification of live births started in 2008. Since 2011, only electronic certificates are used. The use of this system has improved the coverage and timeliness of birth registration, as almost all births occur within the health system (more than 99% since 2008).
- Under-registration of births is believed to be 0% since 2011. Before 2011, birth registration was incomplete: from 1996 to 2005, under-registration of births was about 2.5%. This proportion declined from 2006 onwards, reaching 0% in 2010-2011.
- Data from 1981 to 1993 are by year of registration. Data from 1996 onwards are by year of occurrence of the birth.

Treatment of births by vitality:

- Birth statistics come from civil registration of live births; the stillbirths are therefore not included. Nevertheless, the type of pregnancy (e.g., single, double, etc.) is indicated in the live-birth certificate, even if one or more children from the same pregnancy are born dead. For example, in a double pregnancy where only one of the twins is born alive, the birth certificate of the live-born baby indicates that he/she is a twin. Data on stillbirths by multiplicity are unfortunately not available.
- In the HMBD, the variable Stillbirths = 0 (i.e., stillbirths not included) in Uruguay's input data file, as data from the original sources are counts of babies born alive, by plurality. However, in HMBD's pooled data file, the variable Stillbirths = 2 (i.e., mixed treatment of the stillbirths), as estimates on deliveries by plurality take into consideration the stillbirths, in those cases where there is at least one baby born alive (hence the living twin represents one twin delivery).

Detailed metadata by source

Source: Ministerio de Salud Pública – Instituto Nacional de Estadística (MSP-INE).

References	<u>Detailed sources by period:</u> 1981, 1983, and 1993: Ministerio de Salud Pública. “Nacimientos, Cuadros históricos: Cuadro 23. Nacidos vivos ocurridos y registrados, por edad de la madre y tipo de embarazo” [Births, historical tables: Table 23. Live births (occurred and registered) by mother’s age and type of pregnancy]. Available online: https://uins.msp.gub.uy/natalidad78_88_93.html (Accessed: 03/06/2022). 1982, 1984-1988: Ministerio de Salud Pública. “Nacimientos, Cuadros históricos: Cuadro 24. Nacidos vivos ocurridos y registrados, según sexo, peso al nacer y tipo de embarazo” [Births, historical tables: Table 24. Live births (occurred and registered) by sex, birth weight and type of pregnancy]. Available online: https://uins.msp.gub.uy/natalidad78_88_93.html (Accessed: 03/06/2022). 1996-2020: Ministerio de Salud Pública – Instituto Nacional de Estadística (MSP-INE), “Nacimientos, base de datos” [Births, database], Available online: https://uins.msp.gub.uy/nacimientos.html (Accessed: 03/06/2022).
Period covered	1981-2020 Data are not available for the following years: 1989-1992, 1994, 1995 and 1998.
Code in database	MSP-INE
Coverage and completeness	Birth statistics for Uruguay come from civil registration of live births (<i>Certificado de nacido vivo</i>); the stillbirths are therefore not included. These data are available in the website of the Ministry of Public Health (see References). Data include all births that occurred in Uruguay and are registered in the civil registry (<i>Dirección General del Registro de Estado Civil</i>). Data do not include births registered in Uruguayan consulates abroad. Data up until 2007 come from paper-based certificates of live births. This system resulted in geographic disparities in the timeliness and coverage of birth registration, because public hospitals in Montevideo have civil registry offices, facilitating this process. On the contrary, in the private health sector and in the rest of the country, the parents had the responsibility to register the birth of a child in a civil registry office, up to 10 days after the birth (20 days in rural areas)¹. Gradual implementation of electronic certification of live births started in 2008 (<i>Certificado de nacido vivo electrónico</i>, CNVe). Data from 2008 to 2010 come

¹ Cabella, W., and Peri, A., (2005). *El sistema de estadísticas vitales en Uruguay: elementos para su diagnóstico y propuesta para su mejoramiento*. Estadísticas socio Demográficas en Uruguay: Diagnostico y Propuestas. Facultad de Ciencias Sociales (UdelaR) y Fondo de Población de Naciones Unidas; Ministerio de Educación y Cultura - Dirección General del Registro de Estado Civil (2022). Inscripción de nacimiento. URL: <https://www.gub.uy/tramites/inscripcion-nacimiento> (Accessed: 31/08/2022).

	from both types of certificates (i.e., paper-based and electronic), as this period marks the transition from the former to the latter system. Since 2011, only electronic certificates (CNVe) are used. With CNVe, births are registered in the hospital or health institution of occurrence of the delivery, as information is entered through a web-based tool and sent directly to the Ministry of Public Health.² The use of this system has improved the coverage and timeliness of birth registration in Uruguay, as almost all births occur within the health system (more than 99% since 2008). Under-registration of births is believed to be 0% since 2011. Before 2011, birth registration was incomplete: from 1996 to 2005, under-registration of births was about 2.5%.³ This proportion declined from 2006 onwards, reaching 0% in 2010-2011. Data covering the period from 1981 to 1993 come from aggregate, historical tables on the distribution of live births according to multiplicity and other characteristics such as sex, birth weight, mother's age, and region. These data are by year of registration. The annual number of births in these tables may be slightly under-represented, due to birth under-registration, as discussed above. Unfortunately, there is no available documentation or metadata for data before 1996. Data from 1996 onwards come from anonymized files of individual records. These data were verified and harmonized by a group of experts, who also provided documentation and metadata for these files (see References). In the HMBD, data from these individual records is by year of occurrence of the birth. For this reason, there are discrepancies between HMBD's data and aggregate figures on births published by the Uruguayan Ministry of Public Health for the period 1996-2010, as the latter use the year of registration of the birth. Since 2011, the Ministry publishes tables using the year of occurrence of the birth. In the microdata files, the year of registration is the same as the year of occurrence for most births (97-99% depending on the year). For the other 1-3%, the year of registration is the one immediately after the year of birth. Among the annual microdata files of the births database, only those from 2011 onwards contain a variable that allows distinguishing the type of multiple birth (e.g., double, triple, etc.). In the files for the previous years (1996-2010), there is one variable that allows identifying whether a baby is born from a single or a multiple pregnancy (i.e., without distinguishing twins, triplets, etc.) and another variable that provides the birth order of each baby born alive. Because of this, data on the number of children by multiplicity (columns <i>Twin_children</i>, <i>Triplet_children</i> and <i>Quadruplet_plus_children</i>) provided in Uruguay's 'input
--	---

² Natan, M. (2014). "Hacia un régimen fecundidad tardía? Un análisis de período y cohorte sobre la edad al primer hijo en Uruguay", Facultad de Ciencias Sociales, Universidad de la Republica (UdelaR), URL: https://www.colibri.udelar.edu.uy/jspui/bitstream/20.500.12008/7703/1/TMDEP_NathanMath%C3%ADas.pdf (Accessed: 30/08/2022).

³ Instituto Nacional de Estadística (2014), "Estimaciones y proyecciones de la población de Uruguay: metodología y resultados (revisión 2013). URL: https://www.ine.gub.uy/c/document_library/get_file?uuid=c4d937f9-49e4-4989-b3fc-c6130745233b&groupId=10181 (Accessed: 30/08/2022).

Nathan (2014) applied the same estimate of under-registration (2.5%) to births data for the years 1978-1988 and 1993.

	data' sheet for the period 1996-2010 are approximations, as they are estimated based on the information provided by the two variables mentioned previously.
Definitions and treatment of births by vitality	Birth statistics for Uruguay come from civil registration of live births (<i>Certificado de nacido vivo</i>); the stillbirths are therefore not included. Nevertheless, the type of pregnancy is indicated in the live-birth certificate, even if one or more children from the same pregnancy are born dead. For example, in a double pregnancy were only one of the twins is born alive, the birth certificate of the live-born baby indicates that he/she is a twin (or a baby from a multiple pregnancy for data before 2011, see Coverage and completeness). Data on stillbirths by multiplicity are unfortunately not available. In the HMBD, the variable Stillbirths = 0 (i.e., stillbirths not included) in Uruguay's input data file, as data from the original sources are counts of babies born alive, by plurality. However, in HMBD's pooled data file, the variable Stillbirths = 2 (i.e., mixed treatment of the stillbirths), as estimates on deliveries by plurality take into consideration the stillbirths, in those cases where there is at least one baby born alive (hence the living twin represents one twin delivery). Live births and stillbirths are defined below. Live births: The international definition of a live-birth is used: "Live birth is the complete expulsion or extraction from its mother of a product of conception, irrespective of the duration of pregnancy, which after such separation, breathes or shows any other evidence of life, such as beating of the heart, pulsation of the umbilical cord or definite movement of voluntary muscles, whether or not the umbilical cord has been cut or the placenta is attached; each product of such a birth is considered liveborn (all live-born infants should be registered and counted as such, irrespective of gestational age or whether alive or dead at the time of registration, and if they die at any time following birth they should also be registered and counted as deaths)."⁴ Stillbirths: Death in utero of a fetus with a gestational age of 20 or more weeks and/or weighing more than 500 grams.⁵
Footnotes⁶	1 - Under-registration of births (about 2.5%) until 2005. Then decline from 2006 onwards, reaching 0% in 2011. 2 - For the years concerned by this note (1982 and 1984-1988), the official data sources available do not provide the detailed distribution of multiple births by multiplicity. For this reason, in the HMBD we calculated the number of deliveries by multiplicity for those years as follows:

⁴ United Nations Statistic Division. (2017). "Natality, Concepts and definitions: Live birth", URL: <https://unstats.un.org/unsd/demographic/sconcerns/natality/natmethods.htm> (Accessed: 07/06/2022).

⁵ IMPO – Centro de información oficial. (2019). Decreto N° 276/019: "Reglamentación de la ley 19.628, relativa a la creación del marco normativo para la expedición del certificado de defunción", URL: <https://www.impo.com.uy/bases/decretos/276-2019> (Accessed: 31/08/2022).

⁶ Number indicated in the variable *Footnotes*.

$$D_{(t, \text{twins})} = (B_{(t, \text{multiple})} * w_{(\text{twins})}) / 2,$$

$$D_{(t, \text{triplets})} = (B_{(t, \text{multiple})} * w_{(\text{triplets})}) / 3,$$

$$D_{(t, \text{quadruplets} +)} = (B_{(t, \text{multiple})} * w_{(\text{quadruplets} +)}) / 4.$$

$D_{(t, \text{twins})}$, $D_{(t, \text{triplets})}$, and $D_{(t, \text{quadruplets} +)}$ are the estimated numbers of twin, triplets, and quadruplet + deliveries, respectively. $B_{(t, \text{multiple})}$ is the number of multiple births (children) observed in the year t . This number is multiplied either by $w_{(\text{twins})}$, $w_{(\text{triplets})}$, or $w_{(\text{quadruplets} +)}$, which are the average proportions of twins, triplets, and quadruplets and more children, respectively, among the multiple births. These parameters are constants, as we did not observe trends over time in the proportion of these births among the multiple births, for the years for which the distribution of multiple births by plurality is known. Hence:

$$w_{(\text{twins})} \approx 0.97$$

$$w_{(\text{triplets})} \approx 0.02$$

$$w_{(\text{quadruplets} +)} = 1 - w_{(\text{twins})} - w_{(\text{triplets})}.$$

- 3** - For the years concerned by this note (1993 and 2011-2015), the total number of children is slightly greater than the sum of singletons and multiple children (i.e., *Total_children* > *Singletons* + *Multiple_children*, in the 'input data' file for Uruguay). The difference corresponds to children of unknown multiplicity. The largest difference (8 children) is observed in 2012.

In order to take into consideration the births of unknown plurality in the calculation of the total annual number of deliveries in the HMBD (pooled data file), an approximation to the number of deliveries corresponding to those births is obtained by using the known distributions by plurality as weights, i.e.,:

$$D_{(t, \text{unknown})} = (B_{(t, \text{unknown})} * P_{(t, \text{singleton})}) + ((B_{(t, \text{unknown})} * (1 - P_{(t, \text{singleton})})) / 2),$$

Where $D_{(t, \text{unknown})}$ is the number of deliveries of unknown plurality in the year t , $B_{(t, \text{unknown})}$ is the number of births (children) of unknown plurality in the year t , and $P_{(t, \text{singleton})}$ is the proportion of singleton births (i.e., *Singletons* / *Total_children*) in the year t .

$D_{(t, \text{unknown})}$ is mainly driven by the first term of the above equation, as at least 97% or more births with known plurality are single births (depending on the year). For the other 3% or less (depending on the year) we assume deliveries are twin deliveries – as indicated by the second term of the equation –, as most multiple births are twin births (96% or more depending on the year). Thus, for any given year t , the total number of deliveries is calculated as:

$$D_{(t)} = D_{(t, \text{singletons})} + D_{(t, \text{multiple})} + D_{(t, \text{unknown})},$$

$D_{(t, \text{singletons})}$ and $D_{(t, \text{multiple})}$ are the number of observed single deliveries and the estimated number of multiple deliveries in year t , respectively.

- 4** - From 1996 onwards, the numbers of multiple deliveries by type are slightly underestimated in the HMBD, as data come from individual certificates of

live births. Undercount of multiple deliveries occurs when all babies from a multiple pregnancy are stillborn, as in such cases there is not a single certificate of live birth informing of a multiple delivery (see **Definitions and treatment of births by vitality**). Unfortunately, data on stillbirths by multiplicity are not available.

- 5 - From 1996 to 2010, the number of children by multiplicity in the 'input data' file (columns *Twin_children*, *Triplet_children* and *Quadruplet_plus_children*) are approximations, as they are estimated based on the information provided by two variables: one that allows identifying whether a baby is born from a single or a multiple pregnancy (i.e., without distinguishing twins, triplets, etc.) and another one that provides the birth order of each baby born alive.
- 6 - For the years concerned by this note (1996, 1997, 1999-2003, 2005, 2006, 2008 and 2010), the total number of multiple children is greater than the sum of twins, triplets, and quadruplets and more (i.e., $Multiple_children > Twin_children + Triplet_children + Quadruplet_plus_children$) in the input data file for Uruguay. The difference corresponds to multiple children for whom the specific type of multiple delivery is not available. The largest discrepancy is observed in 2010 (67 children).

In the HMBD, we take these births into account for estimating the number of deliveries by plurality as well as the twinning rate and the multiple rate for the concerned years, as we distributed them proportionally among the types of multiple births based on the known information. For example, the numbers of twin, triplet, and quadruplet + deliveries for those years ($D_{(t, \text{twins})}$, $D_{(t, \text{triplets})}$, and $D_{(t, \text{quadruplets} +)}$, respectively) are calculated as:

$$\begin{aligned} D_{(t, \text{twins})} &= (B_{(t, \text{twins})} + (B_{(t, \text{unknown multiple})} * w_{(twins)})) / 2 \\ D_{(t, \text{triplets})} &= (B_{(t, \text{triplets})} + (B_{(t, \text{unknown multiple})} * w_{(triplets)})) / 3 \\ D_{(t, \text{quadruplets} +)} &= (B_{(t, \text{quadruplets} +)} + (B_{(t, \text{unknown multiple})} * w_{(quadruplets +)})) / 4 \\ D_{(t, \text{multiple})} &= D_{(t, \text{twins})} + D_{(t, \text{triplets})} + D_{(t, \text{quadruplets} +)} \end{aligned}$$

$B_{(t, \text{twins})}$, $B_{(t, \text{triplets})}$, and $B_{(t, \text{quadruplets} +)}$ are the observed numbers of births (children) by plurality, i.e., twins, triplets, and quadruplets and more. $B_{(t, \text{unknown multiple})}$ is the number of multiple children of unknown plurality in the year t . $w_{(t, \text{twins})}$, $w_{(t, \text{triplets})}$, and $w_{(t, \text{quadruplets} +)}$ are weights, as explained in footnote 2. Then, the twinning rate in the year t ($tr_{(t)}$) and the multiple rate in the year t ($mr_{(t)}$) are estimated as:

$$\begin{aligned} tr_{(t)} &= (D_{(t, \text{twins})} / D_{(t)}) * 1000, \\ \text{and} \\ mr_{(t)} &= (D_{(t, \text{multiple})} / D_{(t)}) * 1000, \end{aligned}$$

where $D_{(t, \text{multiple})}$, the estimated number of multiple deliveries, is:

$$D_{(t, \text{multiple})} = D_{(t, \text{twins})} + D_{(t, \text{triplets})} + D_{(t, \text{quadruplets} +)}.$$

$D_{(t)}$, the total number of deliveries in the year t , is calculated as shown in footnote 3.